

Nepal College of Information Technology (NCIT)

Balkumari, Lalitpur
Affiliated to Pokhara University

Department of IT / Computer Engineering

LAB REPORT

Data Science and Analytics

Lab No.	04
Title	Correlation Analysis and Hypothesis Testing
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1 Objectives

After completing this laboratory work, students will be able to:

1. Measure correlation between numerical variables using Pearson, Spearman, and Kendall coefficients.
2. Measure the association between numerical and categorical variables using Point-Biserial Correlation and Correlation Ratio (Eta).
3. Analyze relationships between categorical variables using Chi-Square Test and Cramer's V.
4. Apply hypothesis testing methods such as One-Sample t-Test, Independent t-Test, One-Way ANOVA, and Chi-Square Test.
5. Interpret p-values and draw statistical conclusions from data.

2 Introduction

Correlation analysis and hypothesis testing are important statistical techniques used in data science for discovering relationships among variables and validating assumptions from data.

Correlation analysis measures the strength and direction of relationships between variables. A positive correlation indicates that two variables move together, while a negative correlation indicates that one variable increases as the other decreases.

Hypothesis testing is a statistical procedure used to determine whether an observed effect or relationship is statistically significant or simply due to random chance. These techniques allow researchers and analysts to make data-driven decisions.

In this laboratory exercise, academic records of NCIT students were analyzed to investigate relationships among study habits, attendance, internet usage, faculty, gender, and examination performance.

3 Tools and Libraries Used

Library	Purpose
Pandas	Data loading and manipulation
NumPy	Numerical computations
SciPy	Statistical analysis and hypothesis testing
Matplotlib	Data visualization
Seaborn	Statistical plotting and heatmaps

Table 1: Libraries Used

4 Dataset Description

The dataset contains records of 80 students and includes both numerical and categorical attributes.

Attribute	Type	Description
StudentID	Identifier	Unique student identifier
Gender	Categorical	Male / Female
Faculty	Categorical	Computer, Software, Electronics, Civil
StudyHours	Numerical	Average study hours per day
Attendance	Numerical	Attendance percentage
InternetHours	Numerical	Recreational internet usage per day
ExamScore	Numerical	Final examination score
Result	Categorical	Pass or Fail

Table 2: Dataset Attributes

5 Part A: Correlation Between Numerical Variables

5.1 Objective

To determine the relationship between study hours, attendance, internet usage, and examination scores.

5.2 Methods Used

- Pearson Correlation
- Spearman Correlation
- Kendall Correlation

5.3 Results

Correlation Type	Value
Pearson (StudyHours, ExamScore)	0.512
Spearman (StudyHours, ExamScore)	0.466
Kendall (StudyHours, ExamScore)	0.325
Attendance vs ExamScore	0.324
InternetHours vs ExamScore	-0.373

Table 3: Correlation Results

5.4 Interpretation

The results indicate a moderate positive relationship between study hours and examination scores. Attendance also shows a positive correlation with exam performance, whereas internet usage demonstrates a moderate negative correlation. Therefore, students who study more and attend classes regularly tend to perform better, while excessive recreational internet usage is associated with lower scores.

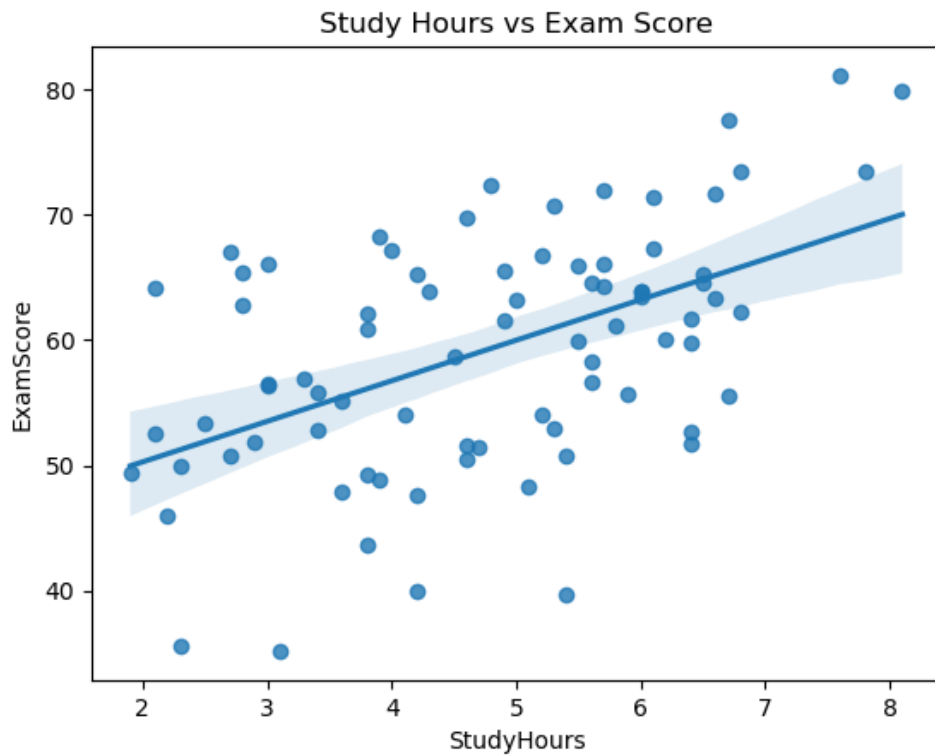


Figure 1: Study Hours vs Exam Score

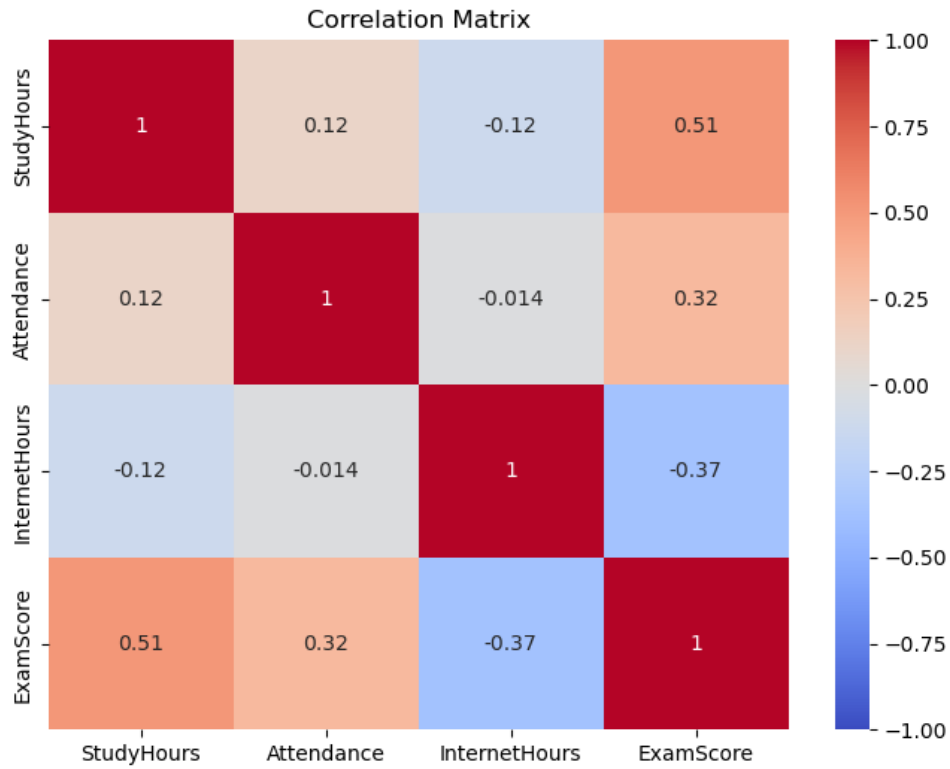


Figure 2: Correlation Matrix Heatmap

6 Part B: Numerical and Categorical Variables

6.1 Point-Biserial Correlation

Point-Biserial Correlation was used to examine the relationship between Gender and ExamScore.

Statistic	Value
Point-Biserial Correlation	-0.219
p-value	0.051

Table 4: Gender vs Exam Score

Interpretation:

The correlation is weak and statistically insignificant because the p-value is greater than 0.05. Therefore, gender does not significantly affect examination scores.

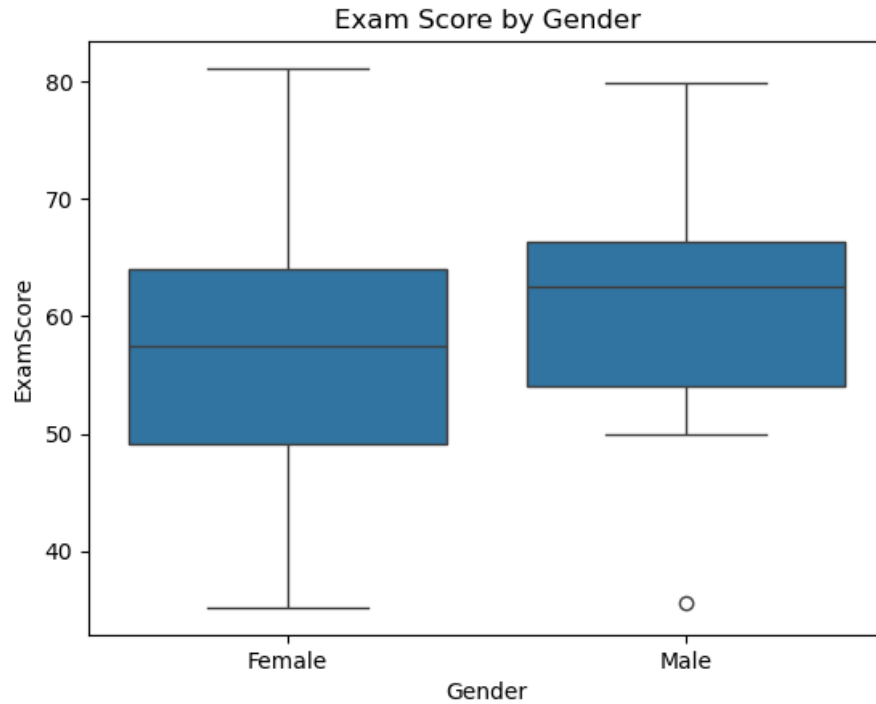


Figure 3: Exam Score Distribution by Gender

6.2 Correlation Ratio (Eta)

Correlation Ratio was used to analyze the relationship between Faculty and ExamScore.

Statistic	Value
Eta (η)	0.511
Eta Squared (η^2)	0.261

Table 5: Faculty vs Exam Score

Interpretation:

Approximately 26% of the variation in examination scores is explained by faculty differences. This indicates a strong association.

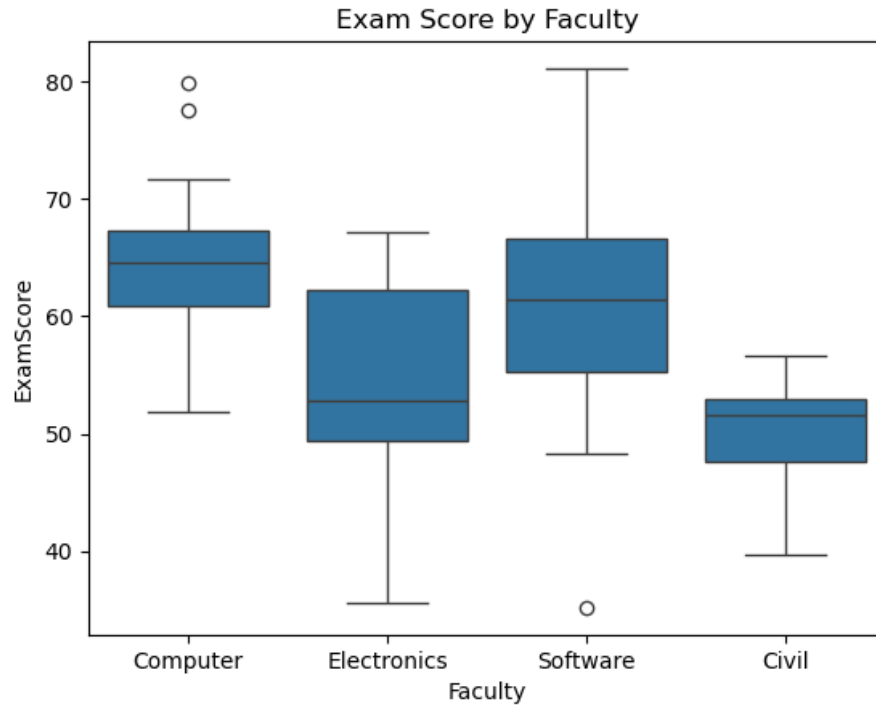


Figure 4: Exam Score by Faculty

7 Part C: Association Between Categorical Variables

7.1 Contingency Table

Faculty	Fail	Pass
Civil	4	5
Computer	0	24
Electronics	6	11
Software	3	27

Table 6: Faculty vs Result

7.2 Chi-Square Test and Cramer's V

Statistic	Value
Chi-Square	15.305
Degrees of Freedom	3
p-value	0.0016
Cramer's V	0.437

Table 7: Categorical Association Results

Interpretation:

Since the p-value is less than 0.05, Faculty and Result are significantly associated. Cramer's V indicates a moderate-to-strong relationship.

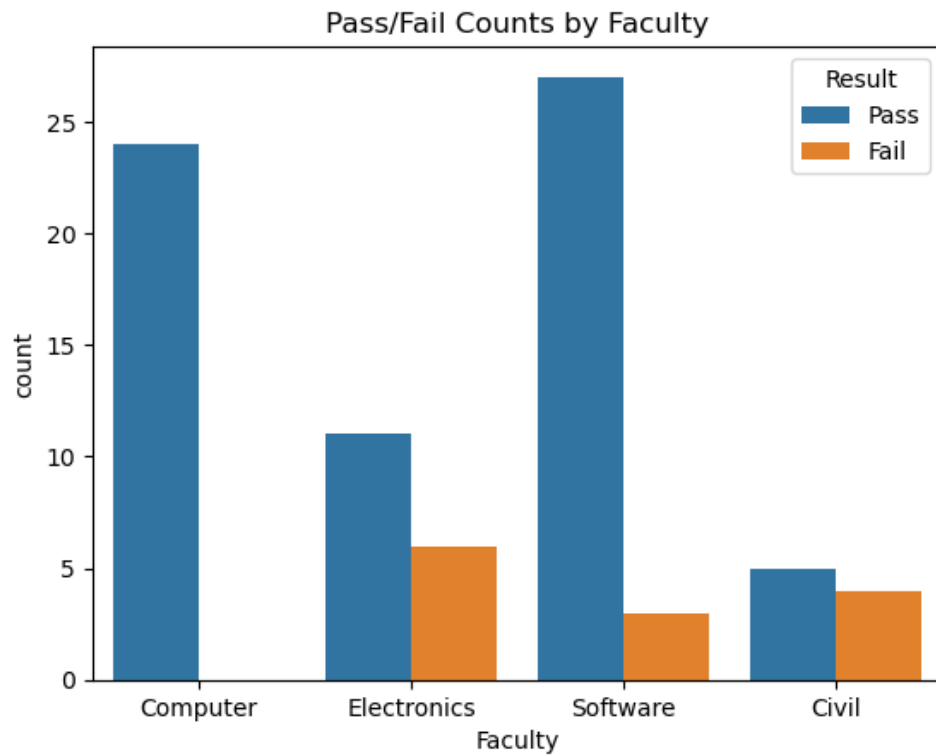


Figure 5: Pass/Fail Counts by Faculty

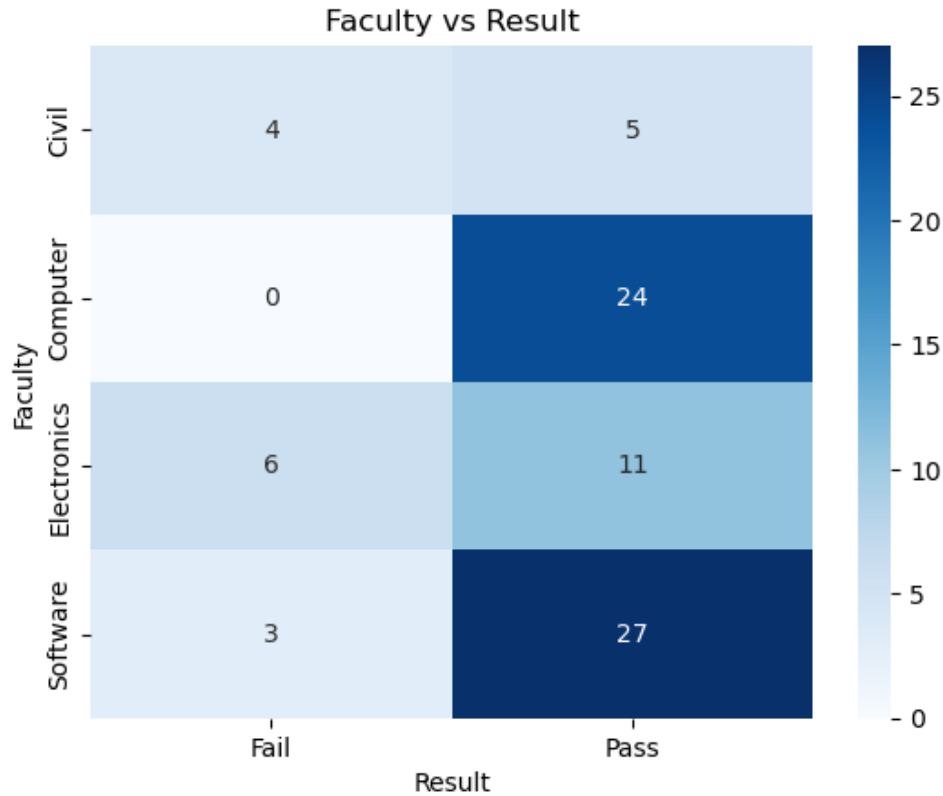


Figure 6: Faculty $\times$ Result Heatmap

8 Part D: Hypothesis Testing

8.1 One-Sample t-Test

Null Hypothesis (H_0): Mean Exam Score = 50

Alternative Hypothesis (H_1): Mean Exam Score $\neq$ 50

Result:

Mean = 59.23, $t = 8.603$, $p = 5.79 \times 10^{-13}$

Decision: Reject H_0

Conclusion: The average exam score differs significantly from 50.

8.2 Independent Two-Sample t-Test

Null Hypothesis (H_0): Male and Female mean scores are equal.

Alternative Hypothesis (H_1): Mean scores differ.

Result:

Female Mean = 56.93

Male Mean = 61.12

$p = 0.0564$

Decision: Fail to Reject H_0

Conclusion: There is no significant difference between male and female examination scores.

8.3 One-Way ANOVA

Null Hypothesis (H_0): All faculty means are equal.

Alternative Hypothesis (H_1): At least one faculty mean differs.

Result:

$$F = 8.957$$

$$p = 3.75 \times 10^{-5}$$

Decision: Reject H_0

Conclusion: At least one faculty differs significantly in examination performance.

8.4 Chi-Square Test of Independence

Null Hypothesis (H_0): Faculty and Result are independent.

Alternative Hypothesis (H_1): Faculty and Result are associated.

Result:

$$\text{Chi-Square} = 15.305$$

$$p = 0.0016$$

Decision: Reject H_0

Conclusion: Faculty significantly influences pass/fail outcomes.

9 Summary of Hypothesis Tests

Test	Decision
One-Sample t-Test	Reject H_0
Two-Sample t-Test	Fail to Reject H_0
One-Way ANOVA	Reject H_0
Chi-Square Test	Reject H_0

Table 8: Summary of Statistical Tests

10 Conclusion

This laboratory successfully demonstrated the application of correlation analysis and hypothesis testing techniques on student academic data. The analysis showed that study hours and attendance positively influence examination performance, while internet usage negatively affects scores. Faculty was found to have a significant impact on academic

performance and pass rates. Gender showed no statistically significant effect on examination scores. The results highlight the importance of statistical analysis in understanding relationships among variables and making informed decisions based on data.